# SREERAM RADHAKRISHNAN

**Dual Degree (B.Tech + M.Tech), Electrical Engineering | Indian Institute of Technology Madras | EE23B075**

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## EDUCATION AND SCHOLASTIC ACHIEVEMENTS

| Program | Institution | %/CGPA | Year |
|---|---|---|---|
| Dual Degree (B.Tech + M.Tech), Electrical Engineering | Indian Institute of Technology Madras | **8.68** | 2028 |
| Class XII (CBSE) | Chettinad Vidyashram, Chennai | 97.2% | 2022 |
| Class X (CBSE) | Atomic Energy Central School, Anupuram | 96.2% | 2020 |

- **Gold Medalist** in the **VLSI** Problem Statement released by **ISRO** at the **14th Inter IIT Tech Meet** (2025).
- **JEE Advanced 2023**: All India Rank **890**; **JEE Main 2023**: All India Rank **747** and **100 percentile** in Physics, out of over 1.2 million students nationwide.

## PUBLICATIONS

- Co-authored, "APEX: A Dual-Sparsity Accelerator for Precise and Efficient SNN Inference", **ISVLSI 2026**. *Other contributions under preparation for submission to IEEE TVLSI.*

## RELEVANT COURSEWORK AND SKILLS

| | | |
|---|---|---|
| Embedded Memory Design | Computer Organization | Secure Processor Microarchitecture |
| Digital IC Design | CAD for VLSI Systems* | Solid State Devices |
| Machine Learning Foundations | Devices and Technologies for AI and Neuromorphic Computing* | ** Ongoing courses* |

**Languages:** SystemVerilog/Verilog, C, C++, Python, Bash, RISC-V, ARM and AVR Assembly

**EDA & Tools:** Synopsys DC, Cadence Xcelium, Genus, Innovus, Tempus, Virtuoso, Calibre, Vivado HLS, KLayout, NGSpice, LTSpice, PyTorch

## TECHNICAL EXPERIENCE

### Agrani Labs

*AI GPU startup, Summer Intern*

May – Jul 2026

*Unified multi-precision transcendental function unit for GPUs | SystemVerilog, Synopsys DC, formal DPV*

- Designed the **RTL** of a unified **multi-precision transcendental function unit** computing multiple functions across floating-point formats with shared hardware.
- Took it through microarchitecture, **synthesis**, **post-synthesis netlist verification** and **formal datapath verification**, delivering a verified block optimized for PPA.

## TECHNICAL PROJECTS AND ACHIEVEMENTS

### Digital Dual-Sparse Neuromorphic Accelerator

*Brainseek Lab, IIT Madras*

Prof. Gopalakrishnan Srinivasan

Jan 2026 – Present

*SystemVerilog, Synopsys DC, Python | ISA design, mixed-precision datapath, dataflow design-space exploration*

- Architecture and microarchitecture of a **digital dual-sparse neuromorphic accelerator** built end-to-end with its software stack.
- Designed a custom **dimension-agnostic ISA** from scratch for the accelerator's sparsity and precision requirements; mapped quantized SNN weights and parameters to a unified, **fully integer-arithmetic mixed-precision datapath**.
- Built a **hardware implementation of the PASC-IF neuron** as a fully combinational datapath with no added latency, letting a converted SNN reach ANN-equivalent accuracy at far fewer timesteps and substantially lower energy. Published as **APEX** at **ISVLSI 2026**; further work under preparation for IEEE TVLSI.
- Ran a systematic **dataflow design-space exploration** for the PE array, deriving a closed-form model of on-chip SRAM traffic, off-chip traffic and storage overhead that unifies multiple candidate dataflows; identified a schedule and array shape that substantially reduces SRAM reads at lower per-PE storage cost, validated on VGG-16, ResNet-18 and AlexNet.

### ASIC Physical Design – ISRO VLSI, Gold Medal

*Inter IIT Tech Meet 14*

Decimation filter of a $\Delta\Sigma$ ADC

Nov – Dec 2025

*Verilog, Xcelium, Genus, Innovus, Calibre, Tempus | SCL 180 nm*

- Implemented the **RTL-to-GDSII flow** with Cadence tools for synthesis, place-and-route and post-layout simulation; **DRC- and LVS-clean** layout.
- **Post-layout STA signoff** in Tempus across SS/FF PVT corners with single-mode and OCV analysis; positive setup and hold slack in all modes.
- Improved PPA through architectural and physical-design choices, validated with back-annotated simulations.

### Custom Layout and RTL-to-GDS in sky130

*EE5311 Digital IC Design*

Jul – Nov 2025

*NGSpice, KLayout | SystemVerilog, STA | SkyWater 130 nm*

- Designed and simulated CMOS logic (16-bit ripple-carry adder, ring oscillator) in **NGSpice** with the sky130 PDK.
- Created physical layouts in **KLayout**; performed **DRC** and **LVS** checks and validated schematic-to-layout consistency.
- Implemented Brent-Kung and Kogge-Stone **carry-lookahead adders** (delay, frequency) and a 16-bit **carry-bypass adder** in SystemVerilog with constraint-based STA to identify the critical path and slack.

### ASR Use-Case Adaptation

*AI4Bharat Lab, IIT Madras*

Prof. Mitesh Khapra

Sep 2024 – Sep 2025

*Python, PyTorch, Pandas, Bash | ASR, low-resource languages*

- Developed an agentic framework for synthetic text generation to adapt **ASR systems** to specialized domains and **low-resource languages**.
- Compared the technique against text-only methods (shallow fusion with external LMs, hotword boosting, LLM rescoring) on ASR performance.

### Project NightVision

*Low-light image enhancement*

May – Dec 2024

*Deep Learning, PyTorch, NumPy | Retinex, attention, ViT, U-Net, GANs*

- Literature review of Retinex theory and attention mechanisms; implemented Vision Transformer, U-Net and GAN architectures from scratch in PyTorch.
- Built a Retinex-based model with iterative illumination refinement and reflectance denoising for low-light enhancement.

## POSITIONS OF RESPONSIBILITY

- **AI Club Coordinator**, Centre for Innovation, IIT Madras (May 2024 – Mar 2025): conducted sessions on machine learning and computer vision for 1000+ students across India and workshops at other colleges on advanced vision architectures; spearheaded an image-captioning project exploring and implementing several architectures.
- **Avanti Mentor**, Avanti Fellows, IIT Madras Chapter (Nov 2023 – Oct 2024): mentored students from Jawahar Navodaya Vidyalaya, Puducherry and Barwani, for JEE Main and Advanced; prepared mock examination papers tailored to diverse competitive examinations.
- **IViL Volunteer**, IIT for Villages, Shaastra & Saarang (Dec 2024 – Jan 2025): managed logistics and finances of the IViL food stalls, empowering rural women with business skills.